

Incentives, Externalities, and Welfare in RLHF Alignment

March 17, 2026

Abstract

Reinforcement learning from human feedback (RLHF) is foundational to modern AI alignment. However, the standard approach to RLHF aggregates feedback from heterogeneous individuals as if it were generated by a single representative agent, which can produce systems that violate basic social welfare criteria such as Pareto efficiency. We develop a formal framework that casts RLHF alignment as a social choice problem. We introduce alignment externalities—the marginal effect of reweighting an individual’s training data on downstream social welfare—and show that they admit an influence-function characterization that geometrically explains when and why anonymous RLHF distorts inferred preferences away from socially efficient outcomes. Our analysis reveals fundamental recoverability constraints: many Pareto-efficient preference parameters cannot be implemented by anonymous protocols without altering how queries—and thus influence—are distributed across individuals.

To-do

This paper

- Clearly state motivation for the anonymous query re-balancing algorithm
- Clean up remaining proofs
- Finalize market algorithm specification
- Analytically characterize convergence conditions / properties
- Run simulations to prove / illustrate algorithm and study additional questions computationally (e.g., convergence rates, stability)

Broader agenda

- Develop statement of broader challenge: AI active learning in social choice settings and costly/constrained preference elicitation
- What are the target application domains where such a method would shine?
- What

1 Introduction

- We first characterize the sub-optimality gap of anonymous anonymous RLHF alignment (fitting to population-level data as if it originated from a single “representative agent,” rather than estimating a utilitarian social welfare function directly).
 - We study the geometry of the optimality gap for RLHF’s learned social welfare function, which can be derived from the first-order condition of the RLHF loss
 - How does this gap distribute across individuals? What types of preferences are systematically over- or under-represented?
 - We do this by defining the “alignment externality” as an influence function connecting training data weighing and deployment welfare. This can be characterized explicitly via the linear preference representation.
 - This approach clarifies how the gap varies as a function of the underlying choice problem, specifically How does alignment on a specific choice problem generalize to (and thereby impact the loss of) other problems?
- We then construct a decentralized mechanism that corrects for the sub-optimality gap of RLHF by
 - Deploy the alignment externalities as Pigouvian prices in a Lindahl market game where individuals choose how to allocate a fixed unit of influence weight across different query types. Show that the resulting equilibrium implements a socially efficient outcome.

Alignment divides into two conceptually distinct learning problems: (A) learning the preferences of a target individual or population, and (B) learning representations that allow those preferences to be effectively encoded. For the purpose of analyzing how alignment aggregates and redistributes welfare across individuals, the impact of RLHF can be approximated by studying an idealized model that solves only the preference-learning problem, with the representational basis held fixed.

Empirically, RLHF updates to a LLM are typically small in magnitude, low in rank, and concentrated in a few layers, which means that the alignment dynamics can be well-approximated by assuming a fixed linear feature map and restricting attention to how various choices affect the learned preference parameters in that representational basis.

Even more specifically, the final layer of a model that makes selections using a softmax (e.g., a LLM) projects a high-dimensional vector representation of a choice problem into utility space. In essence, an LLM or other large model can be viewed as an encoder that converts choice problems into a space of linearly separable utility features. We therefore focus attention on *linear preference representations*—utility functions that take the form of a high-dimensional inner product between choice problems and a preference vector.

1.1 Our Results

- Summary here.

2 Preliminaries

2.1 Individual Choice

Let $\mathcal{X}$ be a set of contexts and $\mathcal{Y}$ a set of possible actions (e.g., choice problems and choices). There are N agents. Each agent n has preferences represented by a utility function $u_n : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$. For a given context and binary set of actions (x, y, y') , we assume the agent selects according to a stochastic choice function with a sigmoid link, as in the Bradley-Terry-Luce (BTL) model

$$p(y \succ y' \mid y, y', x) = \sigma(u_n(y|x) - u_n(y'|x)) \quad (1)$$

where $\sigma(a) = \frac{\exp(a)}{1+\exp(a)}$ and $u_n : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ describes n 's decision utility from each option. Let $w \in \Delta_N$ be an arbitrary N -dimensional vector of social welfare weights. The utilitarian social welfare function is then defined as

$$\bar{u}(y|x) = \sum_n w_n u_n(y|x) \quad (2)$$

Each context-action pair is embedded into a feature space $\phi(x, y) : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^k$ (e.g., the final activation layer of an LLM). The first important assumption is that each agent's utility is linear in this feature space, i.e. can be represented as an inner product

$$u_n(y|x) = \phi(x, y)^\top \theta_n \quad (3)$$

for some latent preference parameter $\theta_n \in \mathbb{R}^k$.

A direct consequence of linearity is that the utilitarian social welfare function can be represented as a weighted average of the individual preference parameters. Specifically, if we define $\bar{\theta} = \sum_n w_n \theta_n$, then

$$\bar{u}(y|x) = \phi(x, y)^\top \bar{\theta} \quad (4)$$

For any stochastic choice rule that is a function of the utility differences (including the Logistic), linearity implies that each contextual choice problem is completely characterized by a vector

$$\alpha(x, y, y') = \phi(x, y) - \phi(x, y') \quad (5)$$

For notational simplicity, we will write $\alpha_z \in \mathbb{R}^k$ to denote the vector that corresponds to a specific choice problem $z \in \mathcal{Z} = \mathcal{X} \times \mathcal{Y} \times \mathcal{Y}$. For expositional simplicity, we assume the set of candidate queries $\mathcal{Z}$ is finite, but this does not substantively affect the conclusions of our analysis.

2.2 Social Welfare & Pareto Efficiency

Let q_{dep} denote the deployment distribution over choices the model will make that affect the entire population. For each individual n , define deployment welfare as the utility induced by the model’s policy

$$W_n(\theta) = \mathbb{E}_{z \sim q_{\text{dep}}} \left[\alpha_z^\top \theta_n \sigma(\alpha_z^\top \theta) \right] \quad (6)$$

Utilitarian social welfare is then defined as the population-weighted version $W(\theta) = \sum_{n=1}^N w_n W_n(\theta)$, where $w \in \Delta_N$ is an arbitrary vector of social welfare weights. The following lemma shows that the utilitarian preference parameter and utilitarian social welfare weights are connected.

The logistic choice function becomes less stochastic as a preference parameter is scaled up (or, equivalently, its temperature is reduced). This mechanically implies that simply scaling a parameter up can increase welfare, which requires that we slightly modified the standard approach to defining Pareto optimality to accommodate limits.

Lemma 1. *Let a preference set $\{\theta_n\}_{n=1}^N$ be given. For any set of social welfare weights $w \in \Delta_N$, we have*

$$\lim_{\beta \rightarrow \infty} W_w(\beta \bar{\theta}_w) = \sup_{\theta \in \mathbb{R}^k} W_w(\theta) \quad (7)$$

Hence, the utilitarian preference parameter $\bar{\theta}_w$ defines a ray that asymptotically attains the maximal achievable social welfare.

Let $C = \text{Conv}(\{\theta_n\}_{n=1}^N)$ denote the convex hull of the population’s preference parameters. Lemma 1 shows that every member of C is an asymptotically welfare-maximizing direction for some set of welfare weights and, conversely, that no parameter outside the set can asymptotically generate strictly more welfare than the best direction in C when scaled. We will therefore refer to $P = \text{cone}(C)$ as the set of “socially efficient” parameters.

3 The Geometry of RLHF Distortion

This section formalizes how anonymous RLHF induces welfare distortions by failing to internalize the social impact of individual training data. We introduce the notion of *alignment externalities*, show that they admit a precise influence-function representation, and derive an explicit closed-form expression for these externalities under anonymous RLHF with logistic choice models.

3.1 Alignment Externalities

In this section, we define the alignment externality, which quantifies the marginal effect of assigning additional RLHF training queries z to agent n on the deployment utility experienced by them and all other agents. Conceptually and mathematically, the alignment externality is an influence function that captures the sensitivity of social welfare to local perturbations of the training data, analogous to those developed for data attribution (see Hammoudeh and Lowd, 2024, for a review).

An RLHF query distribution is a probability measure $q_{\text{train}} \in \Delta_{N \times \mathcal{Z}}$. For each draw $(n, z) \sim q_{\text{train}}$, an observed label is generated from the individual’s choice model $c \mid (n, z) \sim \text{Bernoulli}(\sigma(\alpha_z^\top \theta_n))$. An RLHF procedure then maps the realized choice data into a population-level preference parameter $\theta \in \mathbb{R}^k$. We focus on maximum likelihood estimators, i.e. ones that can be written as

$$\hat{\theta}_{q_{\text{train}}} \in \arg \max_{\theta \in \mathbb{R}^k} \ell(\theta; q_{\text{train}}) \quad (8)$$

for a population log-likelihood

$$\ell(\theta; q_{\text{train}}) = \mathbb{E}_{(n,z) \sim q_{\text{train}}} \mathbb{E}_{c \mid n,z} [\log p_\theta(c \mid z)] \quad (9)$$

Definition 1. For a training source (n, z) , let

$$q_\varepsilon = (1 - \varepsilon) q_{\text{train}} + \varepsilon \delta_{(n,z)} \quad (10)$$

The **alignment externality** generated by agent n training on query z for agent m is

$$\Gamma_{n,z}^m = \left. \frac{d}{d\varepsilon} W_m(\hat{\theta}_{q_\varepsilon}) \right|_{\varepsilon=0} \quad (11)$$

Lemma 2. Under regularity conditions on the MLE (stated in the Appendix), the first-order welfare effect of upweighting (n, z) on agent m satisfies

$$\Gamma_{n,z}^m = \nabla_\theta W_m(\hat{\theta}_{q_{\text{train}}})^\top \mathcal{J}(\hat{\theta}_{q_{\text{train}}}; q_{\text{train}})^{-1} s_{n,z}(\hat{\theta}_{q_{\text{train}}}) \quad (12)$$

where

$$\mathcal{J}(\theta; q_{\text{train}}) = -\nabla_\theta^2 \ell(\theta, q_{\text{train}}) \quad s_{n,z}(\theta) = \mathbb{E}_{c \mid n,z} [\nabla_\theta \log p_\theta(c \mid z)] \quad (13)$$

are the population information matrix and the score contribution of an individual n and problem z , respectively.

While explicit form of the alignment externality depends on the choice of estimator through $\mathcal{J}$, the welfare gradient and score contributions do not depend on the specific estimator used, and are equal to

$$\nabla_\theta W_n(\theta; q_{\text{dep}}) = \mathbb{E}_{z \sim q_{\text{dep}}} \left[(\alpha_z^\top \theta_n) v(\alpha_z^\top \theta) \alpha_z \right] \quad s_{n,z}(\theta) = (\sigma(\alpha_z^\top \theta_n) - \sigma(\alpha_z^\top \theta)) \alpha_z \quad (14)$$

3.2 The Alignment Externality Under Anonymous RLHF

Anonymous RLHF fits a single parameter $\hat{\theta}^{\text{RLHF}}$ to pooled data, ignoring agent identities at training time. The anonymous RLHF objective minimizes the expected binary cross-entropy loss of

choices that are randomly sampled from individuals according to the distribution q_{train} as

$$\begin{aligned}\ell^{\text{RLHF}}(\theta; q_{\text{train}}) &= -\mathbb{E}_{(n,z) \sim q_{\text{train}}} \mathbb{E}_{c|n,z} \left[-c \log \sigma(\alpha_z^\top \theta) - (1-c) \log (1 - \sigma(\alpha_z^\top \theta)) \right] \\ \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} &\in \arg \max_{\theta \in \Theta} \ell^{\text{RLHF}}(\theta; q_{\text{train}})\end{aligned}\tag{15}$$

Solving for the corresponding population information function and combining with (14) yields the following expression for the alignment externality,

$$\Gamma_{n,z}^{\text{RLHF}} = (\sigma(\alpha_z^\top \theta_n) - \sigma(\alpha_z^\top \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}})) (\alpha_z^\top \eta)\tag{16}$$

where $v(t) = \sigma(t)(1 - \sigma(t))$ and

$$\eta(\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}, \bar{\theta}) = \mathbb{E}_{(n,z) \sim q_{\text{train}}} \left[v(\alpha_z^\top \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}) \alpha_z \alpha_z^\top \right]^{-1} \mathbb{E}_{z \sim q_{\text{dep}}} \left[(\alpha_z^\top \bar{\theta}) v(\alpha_z^\top \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}) \alpha_z \right]\tag{17}$$

Note that η is independent of n and z , and acts as a sufficient statistic for the welfare implications of parameter changes across the entire distribution of deployment problems under RLHF.

Thus, anonymous RLHF distorts welfare along directions that are simultaneously (i) systematically misrepresented in the training data and (ii) socially important at deployment.

Proposition 1. Assume $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ is an interior optimum and $\mathcal{J}^{\text{RLHF}}(\hat{\theta}; q_{\text{train}})$ is non-singular. At the anonymous RLHF population optimum $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$, the q_{train} -weighted average alignment externality vanishes in expectation

$$\mathbb{E}_{(n,z) \sim q_{\text{train}}} [\Gamma_{n,z}^{\text{RLHF}}] = 0\tag{18}$$

Proposition 1 implies that anonymous RLHF equalizes marginal welfare effects only in the aggregate, in the sense that under- and over-representation across agents and problems must average out under the realized training mixture. Consequently, distributional misalignment is summarized by the signs of the conditional externalities: if $\Gamma_{n,z}^{\text{RLHF}} > 0$, then marginally upweighting training source (n, z) increases utilitarian welfare, while if $\Gamma_{n,z}^{\text{RLHF}} < 0$, then upweighting (n, z) decreases utilitarian welfare.

3.3 Reweighting Queries to Attain Social Efficiency Under Anonymous RLHF

The foregoing analysis establishes that anonymous RLHF’s social welfare inefficiencies can be formally recast as its failure to fully internalize the externalities induced by query assignment. In essence, applying RLHF to anonymous population-level data generates outcomes that give too much weight to certain individual’s responses, and not enough to others.

The present section asks whether appropriately re-weighting the query distribution can recover socially efficient outcomes, even when the learning procedure itself remains anonymous. The answer is strongly affirmative: in fact, *any* socially efficient outcome can be implemented through a suitable reweighting of training queries. However, this re-weighting must generally be done in a way that conditions on individual identities.

The goal of alignment is to select a preference parameter θ that induces socially desirable deployment behavior. In the present setting, socially efficient parameters lie in the cone generated by the convex hull of individual preference vectors $\{\theta_n\}_{n=1}^N$.

However, anonymous RLHF does not optimize directly over this set. Instead, it estimates a single logistic choice model by fitting to *aggregated choice frequencies*. As a result, not every socially efficient parameter is representable as the limit of anonymous RLHF training, even with infinite data and perfect optimization. Recoverability characterizes this representational constraint. A parameter θ is recoverable if there exists some assignment of training data weights such that the population-level choice frequencies induced by heterogeneous agents are exactly matched by a single logistic model with parameter θ .

If a parameter is not recoverable, then no amount of reweighting anonymous training data can cause RLHF to converge to that parameter. In this sense, recoverability is a necessary condition for social efficiency under anonymous training.

Let $S = \text{span}\{\alpha_z : z \in \text{supp}(q_{\text{train}})\}$ and P_S denote projection onto S . The following definition formalizes the class of parameters that can be exactly recovered by RLHF from anonymous choice data.

Definition 2. A parameter $\theta \in \mathbb{R}^k$ is **individually mix-recoverable** if there exists, for each query $z \in \mathcal{Z}$, a probability vector $\pi(\cdot | z) \in \Delta_N$ such that

$$\sigma(\alpha_z^\top \theta) = \sum_{n=1}^N \pi(n | z) \sigma(\alpha_z^\top \theta_n) \quad \text{for every } z \in \mathcal{Z} \quad (19)$$

Let Θ^{indiv} denote the set of all such parameters. A parameter is, moreover, **anonymously mix-recoverable** if the distribution conditions on z but is equal across individuals, i.e. $\pi(n | z) = \frac{1}{N}$ for every n . Denote the set of such parameters Θ^{unif} , and note that $\Theta^{\text{unif}} \subseteq \Theta^{\text{indiv}}$ for $\rho_n = \frac{1}{N}$.

The set Θ^{indiv} can be directly characterized explicitly using the fact that the logistic link function σ is strictly increasing.

Proposition 2.

1. A parameter $\theta \in \Theta^{\text{indiv}}$ if and only if

$$\alpha_z^\top \theta \in \left[\min_n \alpha_z^\top \theta_n, \max_n \alpha_z^\top \theta_n \right] \quad \text{for every } z \in \mathcal{Z} \quad (20)$$

2. $P_S C \subseteq P_S \Theta^{\text{indiv}}$

Thus, by appropriately reweighting queries on an individual-by-individual level, RLHF can achieve *any* socially efficient outcome, including the egalitarian one. However, doing this reweighting at the individual level is also generally necessary to achieve this flexibility, as the following result shows.

Proposition 3. *If $S \neq \{0\}$, then $P_S \Theta^{\text{unif}}$ is either empty or a singleton.*

We next characterize how the parameter obtained by anonymous RLHF, which effectively computes the M -projection of the choice data onto the single-parameter logistic model family. Letting $\mu_z = \mathbb{E}_{n \sim q_{\text{train}}(n|z)}[\sigma(\alpha_z^\top \theta_n)]$ denote the population-level choice frequency for z , we have

$$\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} \in \arg \min_{\theta \in \mathbb{R}^k} \mathbb{E}_{(z,n) \sim q_{\text{train}}} \left[\text{D}_{\text{KL}} \left(\text{Bern}(\mu_z) \parallel \text{Bern}(\sigma(\alpha_z^\top \theta)) \right) \right] \quad (21)$$

Finding a parameter that fully rationalizes the choice data (in the sense that it achieves zero population asymptotic training loss) is equivalent to discovering an individual-level query mixture that satisfies the definition of individual recoverability.

Proposition 4. *$P_S \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} \in P_S \Theta^{\text{mix}}$ if $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ achieves zero population (asymptotic) training loss.*

(Intuition to develop: add technical conditions for the necessary direction; note that this is not generically possible, thus anonymous RLHF will generically not recover a socially efficient parameter.)

3.4 The Geometry of Anonymous RLHF Distortion

A key question is what drives deviations between $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ and $\bar{\theta}$ (here assuming $S = \mathbb{R}^k$ for simplicity). When the gap is not too large, a Taylor expansion of the score condition gives

$$\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} - \bar{\theta} \approx \mathcal{J}^{\text{RLHF}}(\bar{\theta}; q_{\text{train}})^{-1} \mathbb{E}_{(n,z) \sim q_{\text{train}}} \left[(\mu_z - \sigma(\alpha_z^\top \bar{\theta})) \alpha_z \right] \quad (22)$$

The second (Fisher information) term accounts for the fact that some directions in preference space generate more statistical information than others. The second term shows that the deviation is an average of choice problems, weighted by whether the empirical response frequency over- or under-shoots the egalitarian preference. Define $m(\bar{\theta}) = (\mathcal{J}^{\text{RLHF}}(\bar{\theta}; q_{\text{train}}))^{-1} \nabla_{\theta} W(\bar{\theta})$. We can re-write these in terms of alignment externalities, to see

$$\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} - \bar{\theta} \approx \mathcal{J}^{\text{RLHF}}(\bar{\theta}; q_{\text{train}})^{-1} \mathbb{E}_{(n,z) \sim q_{\text{train}}} \left[\Gamma_{n,z}(\bar{\theta}) \frac{\alpha_z}{\alpha_z^\top m(\bar{\theta})} \right] \quad (23)$$

This expression says:

1. The distortion vector is a weighted average of query directions
2. The weight on each direction is proportional to its alignment externality
3. Directions with large positive externalities are under-weighted
4. Directions with negative externalities are over-weighted

Locally around $\bar{\theta}$, a first-order Taylor approximation of W further gives

$$W_n(\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}) - W(\bar{\theta}) \approx (\theta_n - \bar{\theta}) \mathcal{J}^{\text{RLHF}}(\bar{\theta}; q_{\text{dep}})(\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} - \bar{\theta}) \quad (24)$$

Thus,

$$W_n(\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}) - W(\bar{\theta}) \approx (\theta_n - \bar{\theta}) M \mathbb{E}_{(n,z) \sim q_{\text{train}}} \left[\Gamma_{n,z}(\bar{\theta}) \frac{\alpha_z}{\alpha_z^\top m(\bar{\theta})} \right] \quad (25)$$

where $M = \mathcal{J}^{\text{RLHF}}(\bar{\theta}; q_{\text{dep}}) \mathcal{J}^{\text{RLHF}}(\bar{\theta}; q_{\text{train}})^{-1}$ relates the train and deployment curvature.

Taken together, this shows that (locally) parameters are distorted in query directions that have large alignment externalities, on average, in the training data. Whether an individual benefits or not from this depends on whether their preferences deviate from the population average in the “same” direction as these distortions, weighted by a geometry that reflects differences between the training and deployment distributions, if any.

4 Applications

5 Simulations

Datasets

- <https://huggingface.co/datasets/facebook/community-alignment-dataset>
- <https://huggingface.co/datasets/openai/coval>
- <https://preflib.github.io/PrefLib-Jekyll/dataset/00070>
- <https://preflib.github.io/PrefLib-Jekyll/dataset/00075>
- <https://preflib.github.io/PrefLib-Jekyll/dataset/00032>
- <https://preflib.github.io/PrefLib-Jekyll/dataset/00029>

References

- Noga Alon, Felix Fischer, Ariel Procaccia, and Moshe Tennenholtz. Sum of us: Strategyproof selection from the selectors. In *Proceedings of the 13th conference on theoretical aspects of rationality and knowledge*, pages 101–110, 2011.
- Thomas Kleine Buening, Jiarui Gan, Debmalya Mandal, and Marta Kwiatkowska. Strategyproof reinforcement learning from human feedback. *arXiv preprint arXiv:2503.09561*, 2025.
- Meng Cao, Shuyuan Zhang, Xiao-Wen Chang, and Doina Precup. Scar: Shapley credit assignment for more efficient rlhf. *arXiv preprint arXiv:2505.20417*, 2025.

- Souradip Chakraborty, Jiahao Qiu, Hui Yuan, Alec Koppel, Furong Huang, Dinesh Manocha, Amrit Singh Bedi, and Mengdi Wang. Maxmin-rlhf: Alignment with diverse human preferences. *arXiv preprint arXiv:2402.08925*, 2024.
- Keertana Chidambaram, Karthik Vinay Seetharaman, and Vasilis Syrgkanis. Direct preference optimization with unobserved preference heterogeneity. *arXiv preprint arXiv:2405.15065*, 2024.
- Daniel Fein and Gabriela Aranguiz-Dias. Influence functions for preference dataset pruning. *arXiv preprint arXiv:2507.14344*, 2025.
- Luise Ge, Daniel Halpern, Evi Micha, Ariel D Procaccia, Itai Shapira, Yevgeniy Vorobeychik, and Junlin Wu. Axioms for ai alignment from human feedback. *Advances in Neural Information Processing Systems*, 37:80439–80465, 2024a.
- Luise Ge, Brendan Juba, and Yevgeniy Vorobeychik. Learning linear utility functions from pairwise comparison queries. *arXiv preprint arXiv:2405.02612*, 2024b.
- Luise Ge, Gregory Kehne, and Yevgeniy Vorobeychik. Optimized distortion in linear social choice. *arXiv preprint arXiv:2510.20020*, 2025.
- Paul Gözl, Nika Haghtalab, and Kunhe Yang. Distortion of ai alignment: Does preference optimization optimize for preferences? *arXiv preprint arXiv:2505.23749*, 2025.
- Jerry Green and Jean-Jacques Laffont. Characterization of satisfactory mechanisms for the revelation of preferences for public goods. *Econometrica: Journal of the Econometric Society*, pages 427–438, 1977.
- Daniel Halpern, Evi Micha, Ariel D Procaccia, and Itai Shapira. Pairwise calibrated rewards for pluralistic alignment. *arXiv preprint arXiv:2506.06298*, 2025.
- Zayd Hammoudeh and Daniel Lowd. Training data influence analysis and estimation: A survey. *Machine Learning*, 113(5):2351–2403, 2024.
- Taywon Min, Haeone Lee, Yongchan Kwon, and Kimin Lee. Understanding impact of human feedback via influence functions. *arXiv preprint arXiv:2501.05790*, 2025.
- Rémi Munos, Michal Valko, Daniele Calandriello, Mohammad Gheshlaghi Azar, Mark Rowland, Zhaohan Daniel Guo, Yunhao Tang, Matthieu Geist, Thomas Mesnard, Côme Fiegel, et al. Nash learning from human feedback. In *Forty-first International Conference on Machine Learning*, 2024.
- Chanwoo Park, Mingyang Liu, Dingwen Kong, Kaiqing Zhang, and Asuman Ozdaglar. Rlhf from heterogeneous feedback via personalization and preference aggregation. *arXiv preprint arXiv:2405.00254*, 2024.
- Ariel D Procaccia and Jeffrey S Rosenschein. The distortion of cardinal preferences in voting. In *International Workshop on Cooperative Information Agents*, pages 317–331. Springer, 2006.

Judy Hanwen Shen, Archit Sharma, and Jun Qin. Towards data-centric rlhf: Simple metrics for preference dataset comparison. *arXiv preprint arXiv:2409.09603*, 2024.

Ali Shirali, Arash Nasr-Esfahany, Abdullah Alomar, Parsa Mirtaheri, Rediet Abebe, and Ariel Procaccia. Direct alignment with heterogeneous preferences. *arXiv preprint arXiv:2502.16320*, 2025.

Anand Siththaranjan, Cassidy Laidlaw, and Dylan Hadfield-Menell. Distributional preference learning: Understanding and accounting for hidden context in rlhf. *arXiv preprint arXiv:2312.08358*, 2023.

Zexu Sun, Yiju Guo, Yankai Lin, Xu Chen, Qi Qi, Xing Tang, Ji-Rong Wen, et al. Uncertainty and influence aware reward model refinement for reinforcement learning from human feedback. In *The Thirteenth International Conference on Learning Representations*, 2025.

Appendix

Related Literature

1. Ge et al. (2024a): “Axioms for alignment.” Critically, relative to this and other recent papers, we focus on the distribution of queries and the inter-relationships between them, not distortion on a single query.
2. Gözl et al. (2025): Bounds distortion for an individual choice problem; studies connection between BTL and Borda count
3. Procaccia and Rosenschein (2006): introduced distortion
4. Shirali et al. (2025): Derives the second-order Taylor expansion + variance estimation; also explores individual-level reward modeling
5. Chakraborty et al. (2024): Apply EM on latent preference variables
6. Chidambaram et al. (2024): Cluster annotators and fit policies for each cluster via EM
7. Park et al. (2024): Characterize RLHF convergence for a MARL setup
8. Siththaranjan et al. (2023): Note bias of learned utility with DPO RLHF, advocate “distributional learning”
9. Halpern et al. (2025)
10. Munos et al. (2024): Nash learning from human feedback
11. Ge et al. (2024b, 2025): papers on linear utility functions; learning rates & distortion

12. Green and Laffont (1977): theorem that VCG mechanisms are essentially unique as efficient DSIC mechanisms
13. Buening et al. (2025): strategyproof RLHF
14. Alon et al. (2011): (“sum of us”) strategyproof sampling
15. Min et al. (2025): analyze RLHF via influence functions
16. Fein and Aranguiz-Dias (2025): “Influence Functions for Preference Dataset Pruning”
17. Sun et al. (2025): “Uncertainty and influence aware reward model refinement for reinforcement learning from human feedback”
18. Cao et al. (2025): “SCAR: Shapley Credit Assignment for More Efficient RLHF”
19. Shen et al. (2024): “Towards data-centric rlhf: Simple metrics for preference dataset comparison”

5.1 Proofs

Proof of Lemma 1. The logistic choice function $\sigma(\beta \alpha_z^\top \bar{\theta})$ converges to an indicator $\mathbb{I}(\alpha_z^\top \bar{\theta})$ as the inverse temperature goes to infinity. Hence

$$\lim_{\beta \rightarrow \infty} W(\beta \bar{\theta}) = \mathbb{E}_{z \sim q_{\text{dep}}} \left[\max\{\alpha_z^\top \bar{\theta}, 0\} \right] \quad (26)$$

Next, note that for any α_z , we have $\alpha_z^\top \bar{\theta} \sigma(\alpha_z^\top \theta) \leq \max\{\alpha_z^\top \bar{\theta}, 0\}$. Taking an expectation over both sides yields

$$W(\theta) \leq \mathbb{E}_{z \sim q_{\text{dep}}} \left[\alpha_z^\top \bar{\theta} \sigma(\alpha_z^\top \theta) \right] \leq \mathbb{E}_{z \sim q_{\text{dep}}} \left[\max\{\alpha_z^\top \bar{\theta}, 0\} \right] \quad (27)$$

for all θ , and the result follows. ■

Proof of Lemma 2. First, apply the chain rule

$$\left. \frac{d}{d\varepsilon} W_m(\hat{\theta}(\varepsilon)) \right|_{\varepsilon=0} = \nabla_{\theta} W_m(\hat{\theta}_{q_{\text{train}}})^\top \left. \frac{d\hat{\theta}(\varepsilon)}{d\varepsilon} \right|_{\varepsilon=0} \quad (28)$$

From there,

$$\left. \frac{d\hat{\theta}(\varepsilon)}{d\varepsilon} \right|_{\varepsilon=0} = \mathcal{J}(\hat{\theta}_{q_{\text{train}}}; q_{\text{train}})^{-1} s_{n,z}(\hat{\theta}_{q_{\text{train}}}) \quad (29)$$

is a just standard influence function, and follows from the typical derivation (e.g., Hammoudeh and Lowd, 2024). Combining these two yields the result. ■

Proof of Proposition 2.

Part 1: $\Rightarrow$ Per the definition of $\theta \in \Theta^{\text{indiv}}$, there exists a π such that, for all z ,

$$\sigma(\alpha_z^\top \theta) = \sum_{n=1}^N \pi(n|z) \sigma(\alpha_z^\top \theta_n) \quad (30)$$

As a convex combination of scalars, the right-hand side must lie between its extreme points, so that $\sigma(\alpha_z^\top \theta) \in [\min_n \sigma(\alpha_z^\top \theta_n), \max_n \sigma(\alpha_z^\top \theta_n)]$ for all z . But then, by the strict monotonicity of σ , this implies that $\alpha_z^\top \theta \in [\min_n \alpha_z^\top \theta_n, \max_n \alpha_z^\top \theta_n]$ for all z .

$\Leftarrow$ Let $z \in \mathcal{Z}$ be arbitrary, and suppose that $\sigma(\alpha_z^\top \theta) \in [\min_n \sigma(\alpha_z^\top \theta_n), \max_n \sigma(\alpha_z^\top \theta_n)]$. Let $\underline{n} = \arg \min_n \sigma(\alpha_z^\top \theta_n)$ and $\bar{n} = \arg \max_n \sigma(\alpha_z^\top \theta_n)$. Then, by the monotonicity of the logistic function, there exists a $\pi(\underline{n}|z) \in [0, 1]$ such that

$$\sigma(\alpha_z^\top \theta) = \pi(\underline{n}|z) \sigma(\alpha_z^\top \theta_{\underline{n}}) + \pi(\bar{n}|z) \sigma(\alpha_z^\top \theta_{\bar{n}}) \quad (31)$$

Applying this construction across all $z \in \mathcal{Z}$ yields the result.

Part 2: $P_S C \subseteq P_S \Theta^{\text{indiv}}$: Fix any $\theta \in C$. By the definition of C , there exists $\theta \in \sum_{n=1}^N \lambda_n \theta_n$ for some $\lambda \in \Delta_N$. By the convexity of linear projection, $\alpha_z^\top \theta \in [\min_n \alpha_z^\top \theta_n, \max_n \alpha_z^\top \theta_n]$ for all $z \in \mathcal{Z}$. But then $\sigma(\alpha_z^\top \theta) \in [\min_n \sigma(\alpha_z^\top \theta_n), \max_n \sigma(\alpha_z^\top \theta_n)]$ for all $z \in \mathcal{Z}$. Hence, $\theta \in \Theta^{\text{indiv}}$. ■

Proof of Proposition 3. When $\theta \in \Theta^{\text{unif}}$,

$$\alpha_z^\top \theta = \sigma^{-1} \left(\frac{1}{N} \sum_{n=1}^N \sigma(\alpha_z^\top \theta_n) \right) \quad \forall z \in \mathcal{Z} \quad (32)$$

where the right-hand side is determined entirely by the population $\{\theta_n\}_{n=1}^N$ for each z . If there exists no θ satisfying this system of linear constraints, then $\Theta^{\text{unif}} = \emptyset$ and hence $P_S \Theta^{\text{unif}}$ is empty.

Otherwise, suppose $\theta, \theta' \in \Theta^{\text{unif}}$. They both satisfy the system of linear constraints, so for every $z \in \mathcal{Z}$, $\alpha_z^\top \theta = \alpha_z^\top \theta'$ and hence $\alpha_z^\top (\theta - \theta') = 0$ for $z \in \mathcal{Z}$. This implies $\theta - \theta'$ is orthogonal to $\text{span}\{\alpha_z : z \in \mathcal{Z}\} = S$. Thus all elements of Θ^{unif} have the same projection onto S . Therefore $P_S \Theta^{\text{unif}}$ is at most a singleton. ■

Proof of Proposition ??. If $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ achieves zero population training loss, then every KL term vanishes on the support of q_{train} , hence

$$\sigma(\alpha_z^\top \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}) = \sum_{n=1}^N q_{\text{train}}(n | z) \sigma(\alpha_z^\top \theta_n) \quad \forall z \in \text{supp}(q_{\text{train}}) \quad (33)$$

is exactly the mixture-recovery condition on the identifiable queries. $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ is only identifiable in the subspace S . Applying Proposition 2, this implies that $P_S \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} \in P_S \Theta^{\text{mix}}$. ■

Proof of Proposition 3. When $\theta \in \Theta^{\text{unif}}$,

$$\alpha_z^\top \theta = \sigma^{-1} \left(\frac{1}{N} \sum_{n=1}^N \sigma(\alpha_z^\top \theta_n) \right) \quad \forall z \in \mathcal{Z} \quad (34)$$

where the right-hand side is determined entirely by the population $\{\theta_n\}_{n=1}^N$ for each z . If there exists no θ satisfying this system of linear constraints, then $\Theta^{\text{unif}} = \emptyset$ and hence $P_S \Theta^{\text{unif}}$ is empty.

Otherwise, suppose $\theta, \theta' \in \Theta^{\text{unif}}$. They both satisfy the system of linear constraints, so for every $z \in \mathcal{Z}$, $\alpha_z^\top \theta = \alpha_z^\top \theta'$ and hence $\alpha_z^\top (\theta - \theta') = 0$ for $z \in \mathcal{Z}$. This implies $\theta - \theta'$ is orthogonal to $\text{span}\{\alpha_z : z \in \mathcal{Z}\} = S$. Thus all elements of Θ^{unif} have the same projection onto S . Therefore $P_S \Theta^{\text{unif}}$ is at most a singleton. ■

Proof of Proposition 4. If $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ achieves zero population training loss, then every KL term vanishes on the support of q_{train} , hence

$$\sigma(\alpha_z^\top \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}) = \sum_{n=1}^N q_{\text{train}}(n \mid z) \sigma(\alpha_z^\top \theta_n) \quad \forall z \in \text{supp}(q_{\text{train}}) \quad (35)$$

is exactly the mixture-recovery condition on the identifiable queries. $\hat{\theta}_{q_{\text{train}}}^{\text{RLHF}}$ is only identifiable in the subspace S . Applying Proposition 2, this implies that $P_S \hat{\theta}_{q_{\text{train}}}^{\text{RLHF}} \in P_S \Theta^{\text{mix}}$. ■